

Dual-Arm Box Opening Workflow

Workflow Summary

The box-opening task is implemented as a staged dual-arm manipulation sequence that combines perception, tool handling, cutting, flap opening, object handling, and sorting.

Main Sequence

- Locate the cutting tool with ArUco perception.
- Grasp the tool with the assigned arm and verify feedback state.
- Detect the box and estimate opening keypoints from RGB-D perception.
- Plan and execute the tape-cutting trajectory.
- Use TCP line and circle primitives to open the box flaps.
- Place the tool back into its safe location.
- Flip or lift the box when required by the task state.
- Detect contents and execute grasp-based sorting.

Key Interfaces

- `right/aruco_poses` provides tool and reference marker poses.
- `right/grasppose` provides box-opening keypoint output.
- `left/grasppose` provides planar grasp output.
- `left/arm_control` and `right/arm_control` execute robot-arm commands.
- `left/gripper_control` and `right/gripper_control` execute gripper commands.
- `feedback_states` reports motion and gripper execution status.

Safety Notes

- Validate calibration before running cutting or flap-opening actions.
- Run each arm primitive at reduced speed before the full dual-arm sequence.
- Keep emergency stop access clear during hardware execution.